

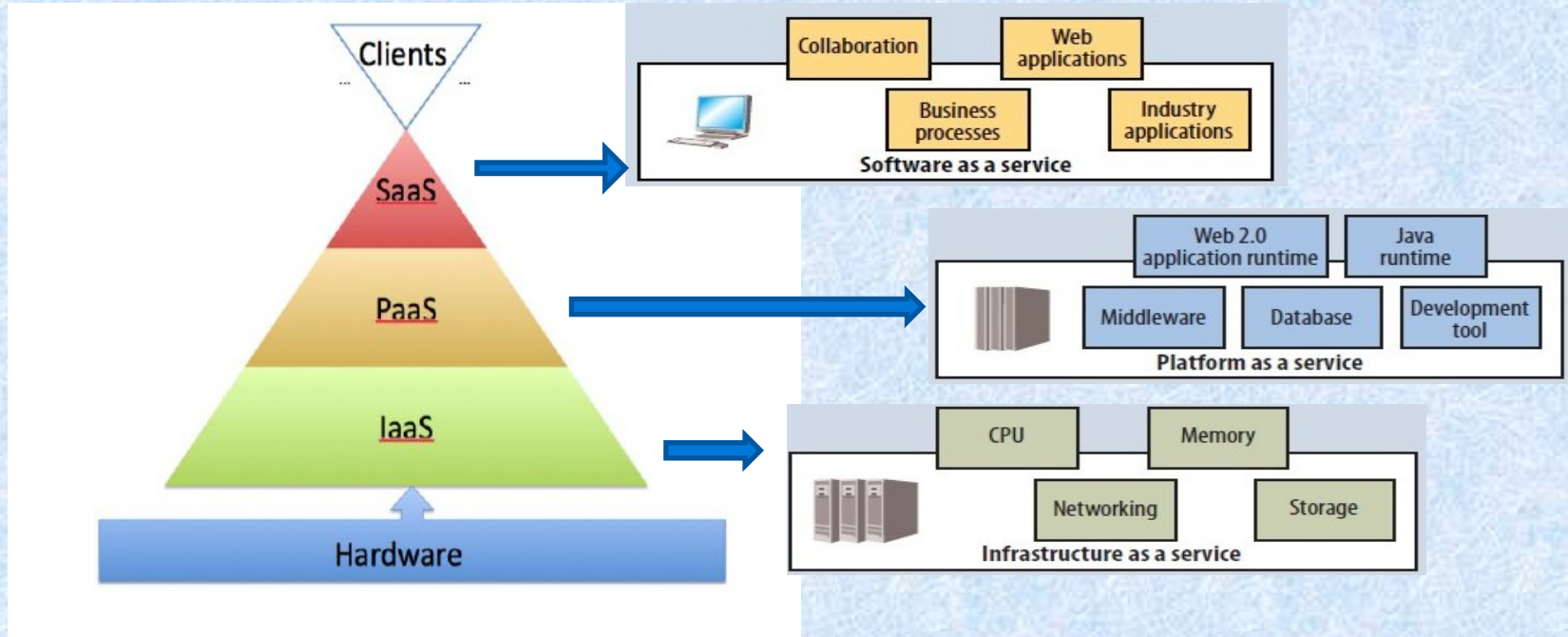


# **Cloud Computing**

**(Cloud Service Models & Cloud Architecture)**

# **CLOUD SERVICE MODELS**

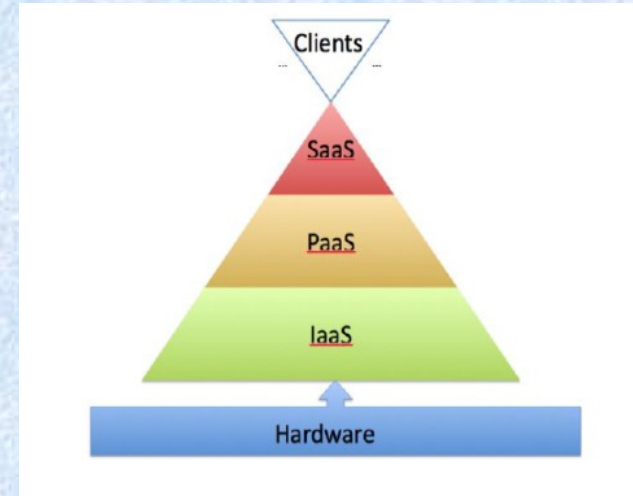
# Overview of Cloud Service Models



# Clients

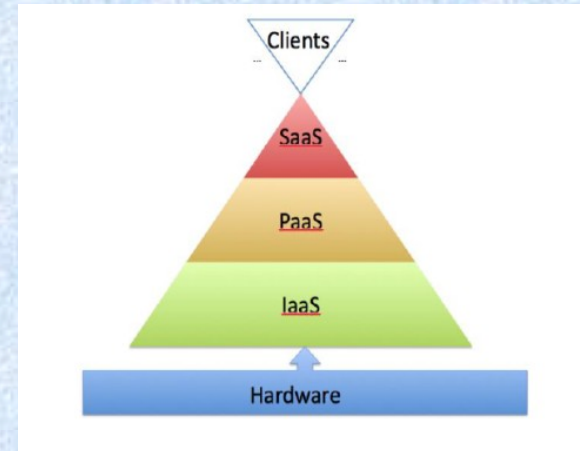
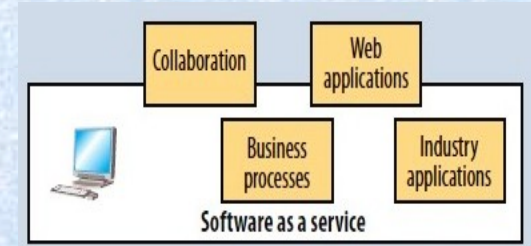


- An **Interface** between Cloud and End User.
- Interact through a predefined set of **rules**.
- Presents the **output/results** in more meaningful way to end users.
- e.g. Web Browser or a thin client application.



# Software-as-a-Service (SaaS)

- Multi-tenant platform for providing applications or software deployed on the Internet, packaged as a unique service for users to consume.
- e.g. Google Docs, Face book, Twitter etc.
- *“SaaS is a software distribution model that gives customers access to applications over the internet, rather than requiring a physical media and custom installation.”*



# Software-as-a-Service (SaaS)

## How does SaaS work?

- SaaS products are centrally hosted by a provider.
- The provider also maintains and updates the software automatically.
- Customers access and use them via the web and mobile browsers.

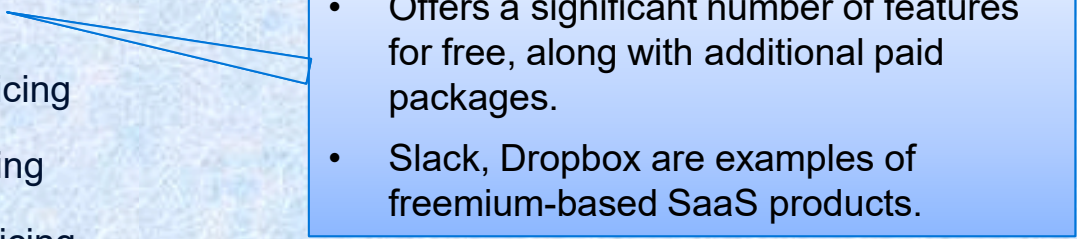
# Software-as-a-Service (SaaS)

## Benefits of SaaS:

- No complex on-site installation
- No custom development & upgradation
- No training needed
- Employees could start using a new tool effectively

# Software-as-a-Service (SaaS)

## SaaS Pricing Models:

- Freemium
  - Flat rate pricing
  - Tiered Pricing
  - Per user pricing
  - Usage based pricing
- 
- Offers a significant number of features for free, along with additional paid packages.
  - Slack, Dropbox are examples of freemium-based SaaS products.

# Software-as-a-Service (SaaS)

## SaaS Pricing Models:

- Freemium
- Flat rate pricing
  - Single product with a standard feature set for a flat rate.
  - Simplest approach for selling SaaS.
  - Basecamp charges a flat rate of \$99 a month for 500 GB of storage.
- Tiered Pricing
- Per user pricing
- Usage based pricing

# Software-as-a-Service (SaaS)

## SaaS Pricing Models:

- Freemium
- Flat rate pricing
- Tiered Pricing 
  - Most common pricing practice.
  - Offer multiple packages.
  - Each package includes a different feature set.
- Per user pricing
- Usage based pricing

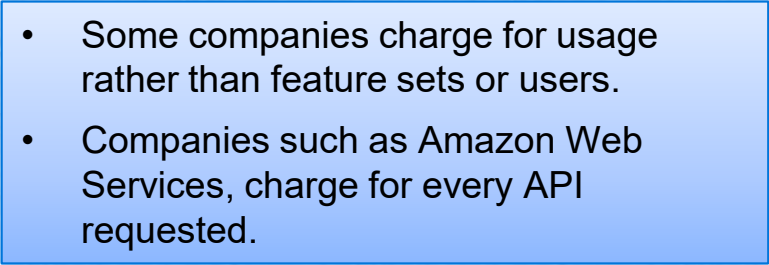
# Software-as-a-Service (SaaS)

## SaaS Pricing Models:

- Freemium
- Flat rate pricing
- Tiered Pricing
- Per user pricing
  - Depending on the number of users.
  - Rather than paying for feature set or flat fee, pay for number of users.
- Usage based pricing

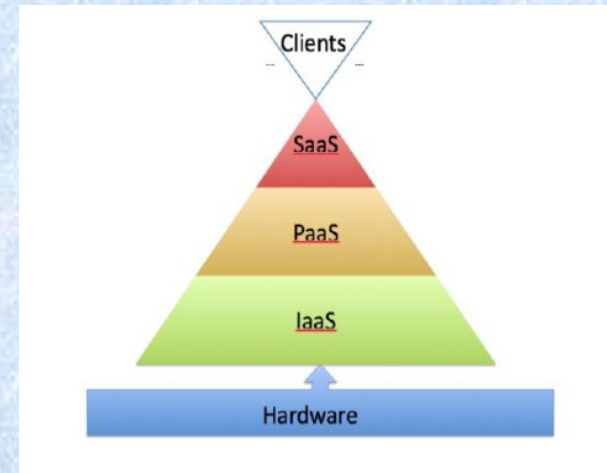
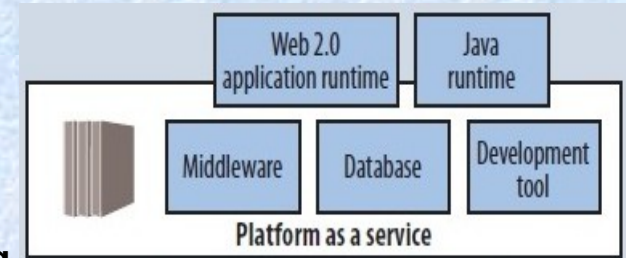
# Software-as-a-Service (SaaS)

## SaaS Pricing Models:

- Freemium
  - Flat rate pricing
  - Tiered Pricing
  - Per user pricing
  - Usage based pricing
- 
- Some companies charge for usage rather than feature sets or users.
  - Companies such as Amazon Web Services, charge for every API requested.

# Platform-as-a-Service (PaaS)

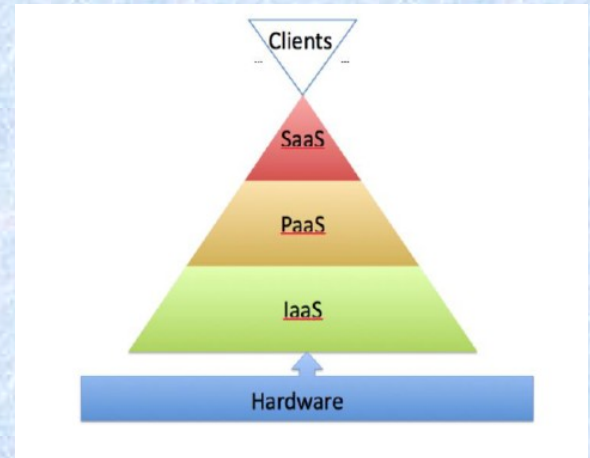
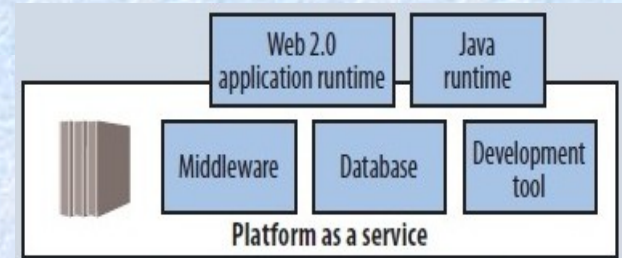
- Create, Host or Deploy a service.
- No need to configure database, servers.
- It is a framework that provides a **unique computing platform** or software stack for applications and services to be developed.
- Application development, collaboration



# Platform-as-a-Service (PaaS)

## How does PaaS work?

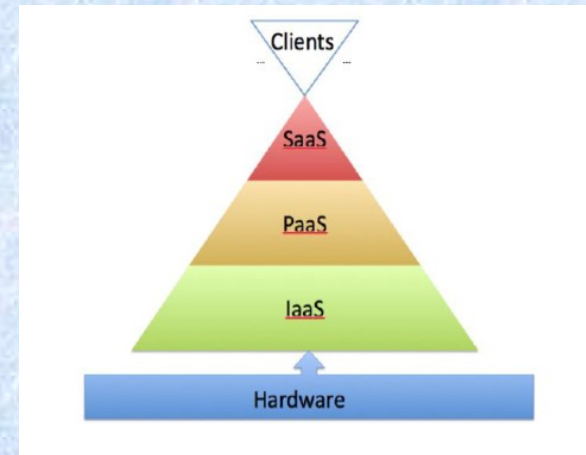
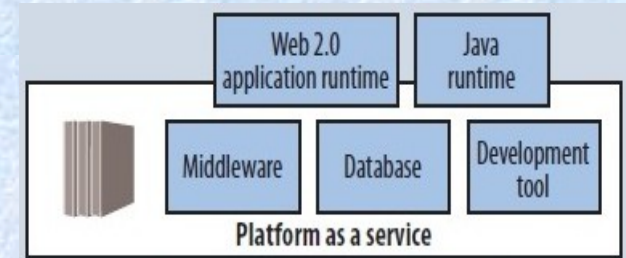
- PaaS does not replace a company's entire IT infrastructure.
- Provided through a cloud service provider's hosted infrastructure.
- PaaS architectures keep their underlying infrastructure hidden from developers and other users.



# Platform-as-a-Service (PaaS)

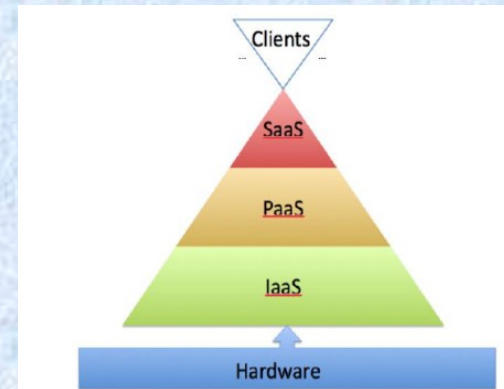
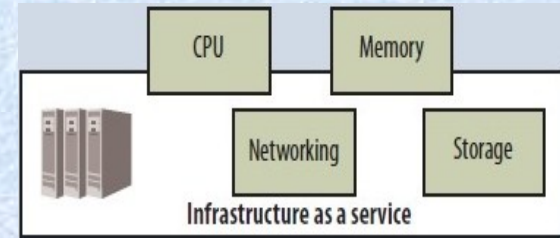
## Benefits of PaaS:

- Simplicity and convenience for users.
- PaaS shifts the responsibility for providing, managing and updating key tools from the internal IT team to the outside PaaS provider.
- Better collaboration



# Infrastructure-as-a-Service (IaaS)

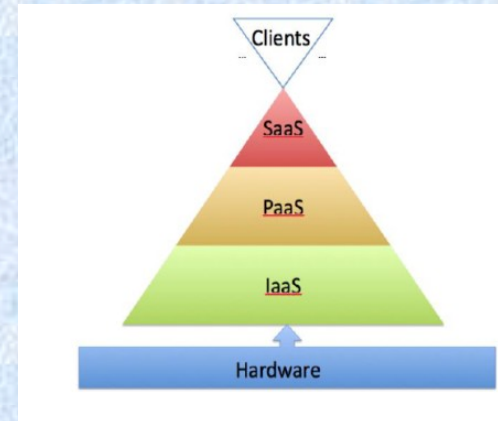
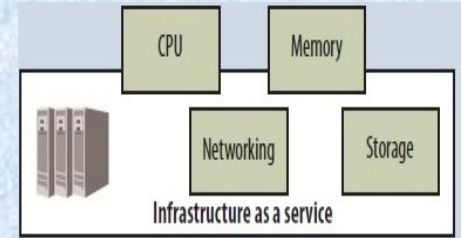
- It is a framework that provides **entire computing resources** through a service.
- This allows users to **rent or buy computing resources** on demand for their own use without needing to operate or manage physical infrastructure.
- e.g. Amazon EC2, Eucalyptus, and Nimbus.
- All IT infrastructure needs: virtual machines, operating system, and networks to storage, support, and hardware



# Infrastructure-as-a-Service (IaaS)

## Benefits of IaaS:

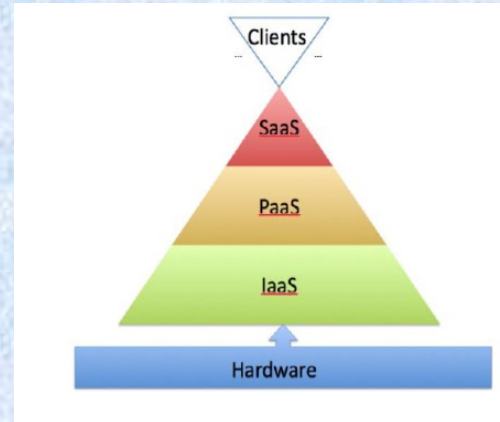
- Best suited for businesses that are new, short-term, or that entail unexpected changes
- Increased Performance
- Security
- Scalability & Flexibility
- Disaster Recovery



# Physical Hardware

## Hardware

- This is the **basis** for all computing infrastructure providing the underlying **set of physical machines** and IT equipment to host the various levels of service.



# Summary

| Category | Characteristics                                                                                                   | Product Type                                                          | Vendors & Products                           |
|----------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------|
| SaaS     | Customers are provided with applications that are accessible anytime and from anywhere                            | Web applications and services (web2.0)                                | Salesforce.com, Google Apps                  |
| PaaS     | Customers are provided with platform for developing applications hosted in the cloud.                             | Programming APIs and frameworks                                       | Google AppEngine, MS Azure, Manjrasoft Aneka |
| IaaS     | Customers are provided with virtualized hardware and storage on top of which they can build their infrastructure. | VM management, infrastructure storage management , network management | Amazon EC2 & S3, GoGrid                      |

# Identifying Cloud Service Models

**Q1.** A company uses Google Docs through a web browser without installing the software locally.

**Q2.** A developer wants a cloud environment where they can develop and deploy an application without managing the underlying servers.

**Q3.** A company rents virtual machines, storage, and networking resources from a cloud provider.

**Q4.** Which service model provides applications directly to end users?

**Q5.** Which service model provides a platform for application development?

**Q6.** Which service model provides virtualized computing and storage resources?

# Identifying Cloud Service Models

**Q7.** What makes SaaS different from traditional software installation?

**Q8.** Why does SaaS reduce the need for on-site installation and software upgrades?

**Q9.** A university wants to provide an online collaboration tool to 500 students. Why might SaaS be a suitable choice?

**Q10.** Which SaaS pricing model would you choose for a small startup: **freemium, flat-rate, tiered, per-user, or usage-based**? Why?

# SaaS Pricing Models

**Scenario: Company A vs Company B**

**Company A has 10 employees. Company B has 500 employees. Both need the same cloud software.**

**Q1.** Which pricing model would potentially make the most sense for each company?

**Q2.** What are the advantages and disadvantages of **per-user pricing**?

**Q3.** When might **flat-rate pricing** be more attractive?

**Q4.** When would **usage-based pricing** be appropriate?

**Q5.** Design your own SaaS pricing model for a student collaboration application.

# SaaS vs PaaS vs IaaS

| Question                                          | SaaS | PaaS | IaaS |
|---------------------------------------------------|------|------|------|
| What does the customer receive?                   |      |      |      |
| Who is the main user?                             |      |      |      |
| Is application development a main purpose?        |      |      |      |
| Does the customer manage physical infrastructure? |      |      |      |
| Example from the slides                           |      |      |      |

# SaaS vs PaaS vs IaaS

| Question                                                 | SaaS                  | PaaS                        | IaaS                                                     |
|----------------------------------------------------------|-----------------------|-----------------------------|----------------------------------------------------------|
| <b>What does the customer receive?</b>                   | Applications/software | Development platform        | Computing infrastructure                                 |
| <b>Who is the main user?</b>                             | End users             | Developers                  | IT/infrastructure users                                  |
| <b>Is application development a main purpose?</b>        | No                    | Yes                         | Not directly                                             |
| <b>Does the customer manage physical infrastructure?</b> | No                    | No                          | No – physical infrastructure is provided by the provider |
| <b>Example from the slides</b>                           | Google Apps           | Google AppEngine / MS Azure | Amazon EC2 & S3                                          |

# Scenario Challenge

## Scenario A

A startup wants to develop and deploy a new web application but does not want to manage servers.

**Q1. Which cloud service model should they choose?**

**Q2. Why?**

**Q3. What responsibilities would the cloud provider handle?**

**Q4. What would the customer still need to manage?**

**Q5. Present your decision to the class in one minute.**

# Cloud Architecture

# What is cloud architecture?

- Cloud architecture is a key element of building in the cloud.
- It refers to the layout and connects all the necessary components and technologies required for cloud computing.



# Cloud architecture

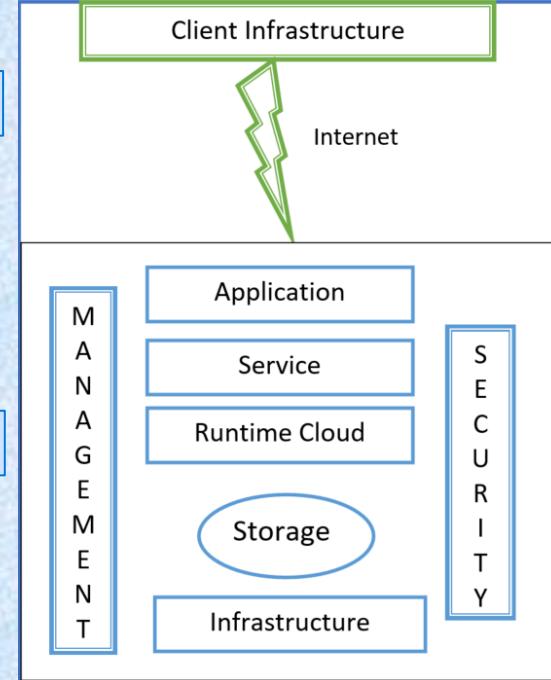
- Cloud architecture refers to how various cloud technology components, such as hardware, virtual resources, software capabilities, and virtual network systems interact and connect to create cloud computing environments.
- It acts as a blueprint that defines the best way to strategically combine resources to build a cloud environment for a specific business need.

# Cloud architecture components

- Frontend
- Backend
- Delivery model
- Network
  - Internet
  - Intranet
  - Inter-cloud

Front End

Back End



# Cloud architecture components

- A frontend platform
- A backend platform
- A cloud-based delivery model
- A network
  - Internet
  - Intranet
  - Intercloud

- Contain the client infrastructure: user interfaces, client-side applications, and the client device or network that enables users to interact with and access cloud computing services.
- GUI

# Cloud architecture components

- A frontend platform
- A backend platform
- A cloud-based delivery model
- A network
  - Internet
  - Intranet
  - Intercloud

- Refers to the cloud architecture components that make up the cloud itself.
- Computing resources, storage, security mechanisms, management, and more.

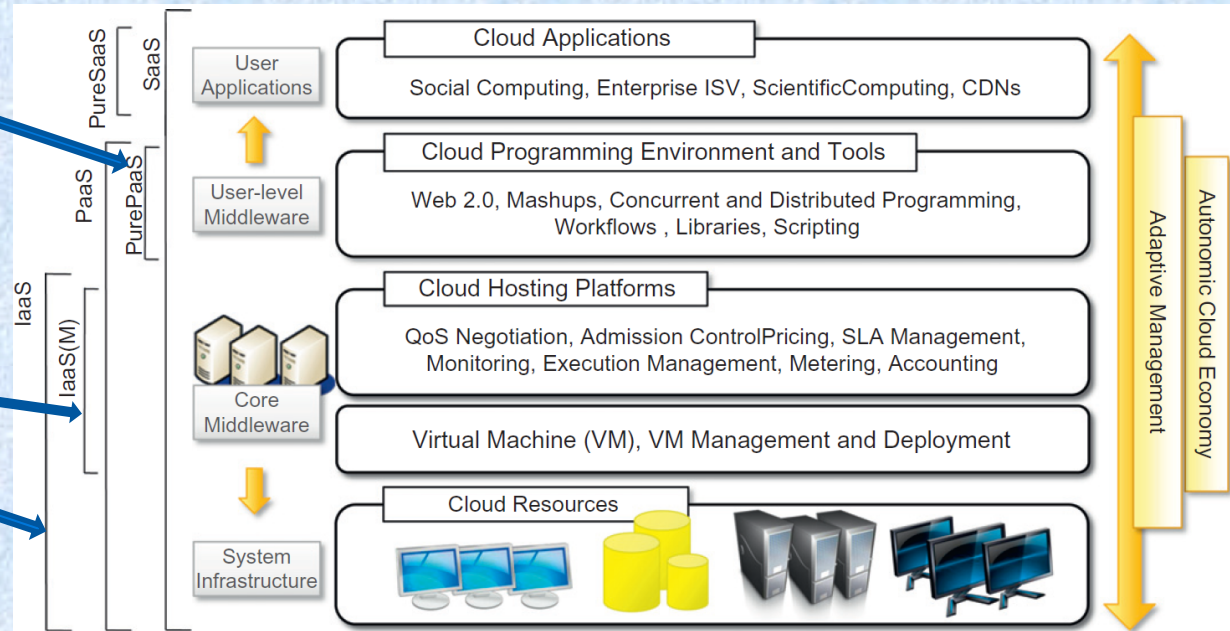
- Application
- Service
- Storage
- Runtime cloud
- Infrastructure
- Management
- Security

# The cloud computing architecture

Pure PaaS, only the user-level middleware is offered, has to be complemented with a virtual or physical infrastructure.

Only the management layer

Provide both the management layer and the physical infrastructure



# Cloud architecture layers

- A simpler way of understanding how cloud architecture works is to think of all these components as various layers placed on top of each other to create a cloud platform.
- Here are the basic cloud architecture layers:
  - **Hardware**
  - **Virtualization**
  - **Application and service**

# Cloud architecture layers

- A simpler way of understanding how cloud architecture works is to think of all these components as various layers placed on top of each other to create a cloud platform.
- Here are the basic cloud architecture layers:
  - **Hardware:** The servers, storage, network devices, and other hardware that power the cloud.
  - **Virtualization**
  - **Application and service**

# Cloud architecture layers

- A simpler way of understanding how cloud architecture works is to think of all these components as various layers placed on top of each other to create a cloud platform.
- Here are the basic cloud architecture layers:
  - **Hardware**
  - **Virtualization:** An abstraction layer that creates a virtual representation of physical computing and storage resources. This allows multiple applications to use the same resources.
  - **Application and service**

# Cloud architecture layers

- A simpler way of understanding how cloud architecture works is to think of all these components as various layers placed on top of each other to create a cloud platform.
- Here are the basic cloud architecture layers:
  - **Hardware**
  - **Virtualization**
  - **Application and service:** This layer coordinates and supports requests from the frontend user interface, offering different services based on the cloud service model, from resource allocation to application development tools to web-based applications.

# Summary

- Cloud Computing Architecture is divided into two parts:
  - front-end and back-end.
- Front-end and back-end communicate via a network or internet.
- Backend refers to the cloud itself which is used by the service provider.
- It contains the resources as well as manages the resources and provides security mechanisms.

**THANK YOU**